

**B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER  
EXAMINATION 2025**

**DIGITAL SIGNAL PROCESSING**

Full Marks 100

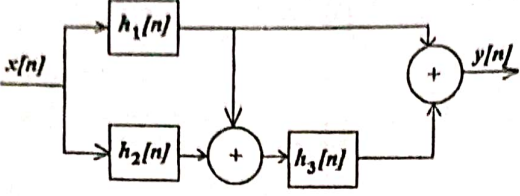
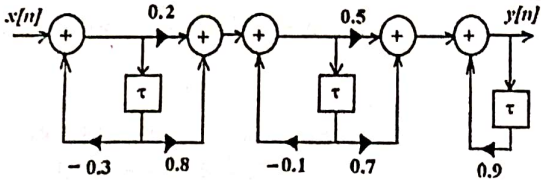
Time: Three hours

(50 marks for each part)

Use a separate Answer-Script for each part

| Question No. | PART- I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|              | <p align="center">Answer any <b>THREE</b> questions<br/>Two marks reserved for well organized answers and neatness</p>                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 1. (a)       | <p>A signal <math>x(t) = \sin^2(20\pi t)\sin(10\pi t)</math>, where <math>t</math> is in second, is uniformly sampled .</p> <p>(i) Determine the Nyquist rate for the signal.</p> <p>(ii) If the sampling frequency is 18 Hz, and the sampled signal is then converted back to analog signal by ideal low-pass filtering with a cutoff frequency of 9 Hz. Determine the frequencies of the harmonic components present in the reconstructed analog signal. Derive only the specific expression(s) relevant to this problem.</p> | 10    |
| 1 (b)        | <p>From the basic definition, determine the Z-transform <math>X(z)</math> of the sequence</p> $x[n] = a^{ n } ; \text{ where }  a  < 1 .$ <p>Express <math>X(z)</math> as a rational function.<br/>State clearly the region of convergence (ROC) of the Z-transform, and give the pole-zero plot.</p>                                                                                                                                                                                                                           | 6     |
| 1 (b)        | <p align="center"><b>OR</b></p> <p>From the basic definition, determine the Z-transform <math>X(z)</math> of the sequence</p> $x[n] = e^{-an} \cos(\omega_0 n) ; \text{ where } a > 0 .$ <p>Express <math>X(z)</math> as a rational function.<br/>State clearly the region of convergence (ROC) of the Z-transform, and give the pole-zero plot.</p>                                                                                                                                                                            | 6     |
| 2. (a)       | <p>Determine the impulse response of the complete system shown in Fig. [A], where the subsystems have impulse responses</p> $h_1[n] = \left(\frac{1}{3}\right)^n u[n] , h_2[n] = \left(\frac{1}{6}\right)^n u[n] \text{ and } h_3[n] = \left(\frac{1}{6}\right)^n u[n].$                                                                                                                                                                                                                                                        |       |

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| Question No. | PART I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|              |  <p style="text-align: center;">Fig. [A]</p>                                                                                                                                                                                                                                                                                                                                                                                                                 | 8     |
| 2(a)         | <p>Also obtain the difference equation relating the output and the input sequences.</p> <p style="text-align: center;">OR</p> <p>Two DTLTI systems have impulse response sequences<br/> <math>g[n] = \{-2, 1, \underset{\uparrow}{8}, -5, 2, 3\}</math> and <math>h[n] = \{-5, 3, 2, \underset{\uparrow}{-6}, 2, -1, 4\}</math></p> <p>Performing computation in time domain, find out the impulse response of the cascade combination of the two systems.</p>                                                                                 | 8     |
| 2. (b)       | <p>The Z-transfer function of a discrete-time linear time-invariant system has a pair of zeros at the origin and another zero at <math>z = -0.8</math>. There is a pair of poles at <math>z = -0.1 \pm j 0.9</math> and another pole at <math>z = 0.7</math>. The DC gain of the system is 6. Obtain the transfer function of the system. Indicate all possible ROCs of the transfer function, and in each case comment on the causality and the stability of the system. Give necessary explanations.</p>                                     | 8     |
| 3. (a)       | <p>A particular structure for realizing an IIR filter is shown in Fig. [B]. Name the structure. Obtain the set of difference equations describing the filter algorithm. Derive the Z-transfer function of the filter. Also determine and draw the Direct form-II and the parallel realizations (using 1<sup>st</sup> order subsystems) of the filter. Give all the relevant sets of difference equations.</p>  <p style="text-align: center;">Fig. [B]</p> | 10    |
| 3.(b)        | <p>How does the left-half of the s-plane map on to the z-plane? Explain.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 6     |

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| Question No. | PART I                                                                                                                                                                                                                                                                                                                                 | Marks |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4.(a)        | Design a low pass digital Butterworth filter with a dc gain of 0 dB, an attenuation of 3 dB at 1620 Hz and an attenuation of at least 10 dB at 3240 Hz. Use impulse invariant transformation. Consider a sampling frequency of 7 kHz.<br><br>Obtain the difference equation relating the output and the input sequences of the filter. | 10    |
| 4.(b)        | Explain the problem of 'Characteristic Aliasing' inherent in the Impulse Invariant Transformation Technique. Under what conditions is the effect of this problem be minimal?                                                                                                                                                           | 6     |
| 5.           | Write short notes on any two of the following.                                                                                                                                                                                                                                                                                         |       |
| (a)          | Frequency Spectra of uniformly sampled signals.                                                                                                                                                                                                                                                                                        | 8+8   |
| (b)          | Recursive and non-recursive DTLTI systems.                                                                                                                                                                                                                                                                                             |       |
| (c)          | Representing uniformly sampled signals by train of scaled impulses.                                                                                                                                                                                                                                                                    |       |
| (d)          | Designing digital IIR filters by bilinear transform with frequency prewarping.                                                                                                                                                                                                                                                         |       |

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**B. E. ELECTRICAL ENGINEERING 2<sup>ND</sup> YEAR 2<sup>ND</sup> SEMESTER EXAMINATION, 2025**

**SUBJECT: - DIGITAL SIGNAL PROCESSING**

Time: Three hours

**Full Marks 100**  
**(50 marks for each part)**

Use a separate Answer-Script for each part

| No. of Questions | PART II                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                     |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
|                  | <i>Answer all the questions.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                           |
| 1.               | Derive the expression of DFT of a sequence of length $N$ . Hence show that the number of time-samples is equal to the number of frequency samples.<br><br>OR<br><br>a) Prove that the algorithm of FFT reduces the computational burden of the processor being used to implement DFT operation. Estimate the savings in computational burden.<br><br>b) "If $W$ is the Twiddle Factor used to perform $N$ -point DFT operation, then always $W^{pN/2} = (-1)^p$ , $p$ being an integer." – Comment whether the statement is true or false. Justify in favour of your comment.<br><br>(CO1-K1)                                                                                                                                                                                                                                                                                           | 10<br><br>6<br>4          |
| 2.               | (a) Show that the magnitude spectrum and phase spectrum of a linear phase digital filter have even and odd symmetry with respect to some folding frequency.<br>(b) What is the relation between this folding frequency and sampling frequency?<br>(c) "A distortion-less filter has constant phase delay in its pass band region." – Comment whether the statement is true or false. Justify in favour of your comment.<br><br>OR<br><br>(a) What are the problems one experiences while designing a brick-wall type FIR digital filter? What measures can you take to overcome the problem? Can the desired frequency response be achieved with such measure? Comment appropriately.<br>(b) Draw the tapped delay structure of an $M$ -tap causal FIR digital filter. Comment on the amount of computational load imposed on the processor to implement the structure.<br><br>(CO2-K2) | 5<br>2<br>3<br><br>6<br>4 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3. | <p>(a) Derive an expression of the frequency response of causal FIR digital filter with symmetric impulse response.</p> <p>(b) How is the group delay of the filter related to the number of coefficients of the filter?</p> <p>(c) Show that the magnitude response and phase response of the filter has even and odd symmetry, respectively.</p>                                                                                                                                                                                                                                                                              | 5<br>2<br>3 |
| OR |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |
|    | <p>(a) State and illustrate with suitable schematic diagram two applications of DFT in the domain of signal processing.</p> <p>(b) A two-input two-output butterfly is used to implement DFT following in-place algorithm. What should be the sequence of mathematical operations needed to realize the butterfly?</p> <p>(CO3-K3)</p>                                                                                                                                                                                                                                                                                          | 7<br>3      |
| 4. | <p>Answer any ONE: (CO4-K4)</p> <p>a) A 7-tap causal linear-phase FIR brick-wall type low-pass filter is designed with a pass band gain of 10 and a cut off frequency of 500 Hz. The sampling frequency is chosen as 2kHz. Determine the filter coefficients and apply Hann window for smoothing filter coefficients. Realize the filter with the pictorial representation of impulse response and tapped delay structure.</p> <p>b) Derive the frequency response of a 7-tap causal Hamming window used over a sampling frequency of 1kHz. What is the approximate width of the main lobe of its frequency characteristic?</p> | 1x10        |
| 5) | <p>Write a short note on any ONE of the following: (CO5-K5)</p> <p>(a) Circular Convolution and its application</p> <p>(b) Distortion-less transmission of signal using FIR digital filter</p>                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1x10        |